

Ch2. Entropy, Relative Entropy, and mutual information

2.1 Entropy

↳ Measure of the uncertainty of a random variable.

X : a discrete random variable.
 $x \in \mathcal{X} \text{ (Y)}$
 $p(x) = \Pr(X=x)$.

Def Entropy of X : $H(X) := - \sum_{x \in \mathcal{X}} p(x) \log p(x)$.

$$* p(x)=0 \Rightarrow \boxed{\lim_{y \rightarrow 0^+} y \log y = 0}$$

↳ $(0 \log 0 = 0)$.

Lemma $H(X) \geq 0$.

$$p(x) \leq 1.$$

$$(\text{pf}) \quad H(X) = - \sum_{x \in \mathcal{X}} p(x) \log p(x) = \sum_{x \in \mathcal{X}} \underbrace{p(x) \log \frac{1}{p(x)}}_{\geq 0} \quad \square$$

Ex. $X = \begin{cases} 1 & \text{with prob } p \\ 0 & \text{" } 1-p \end{cases}$

$$\Rightarrow H(X) = \boxed{-p \log p - (1-p) \log (1-p)} =: H(p).$$

2.2. Joint Entropy & Conditional Entropy.

Ref ~~Joint Entropy~~

$$H(X, Y) = - \sum_{\substack{x \in \mathcal{X} \\ y \in \mathcal{Y}}} p(x, y) \log p(x, y)$$

$$* p(x, y) = \Pr(X=x, Y=y)$$

$$\bullet H(Y|X) = \sum_{x \in \mathcal{X}} p(x) H(Y|X=x),$$

$$= \sum_{x \in \mathcal{X}} p(x) \left(- \sum_{y \in \mathcal{Y}} p(y|x) \log p(y|x) \right)$$

$$* p(y|x) : \text{conditional probability}$$

$$p(y|x) = \frac{p(x, y)}{p(x)}$$

$$(Ex) \textcircled{1} X=Y \sim p(x).$$

$$\Rightarrow H(Y|X) = \sum_{x \in \mathcal{X}} p(x) \left(- \sum_{y \in \mathcal{Y}} p(y|x) \log p(y|x) \right)$$

$$p(y|x) = \begin{cases} 1 & x=y \\ 0 & x \neq y \end{cases} = \delta_{xy}$$

$x = X$
 $\Rightarrow Y = x$

$$\delta_{xy} = 0 \sim 1$$

$$= \sum_{x \in \mathcal{X}} p(x) - \sum_{y \in \mathcal{Y}} \delta_{xy} \log \delta_{xy} = 0$$

12

② X, Y are Independent.

$$p(x, y) = p(x) p(y)$$

$$\frac{H(Y)}{11}$$

$$H(Y|X) = - \sum_{x,y} p(x,y) \log p(y) = - \sum_{x,y} p(y) \log p(y)$$

Thus $H(X, Y) = H(X) + H(Y|X)$.

(p/r) $H(X) + H(Y|X)$

$$= - \sum_x p(x) \log p(x) - \sum_{x,y} p(x,y) \log p(y|x)$$

$$= - \sum_{x,y} p(x,y) \log p(x) - \sum_{x,y} p(x,y) \log p(y|x)$$

$$= - \sum_{x,y} p(x,y) \log p(x,y) = H(X, Y)$$

Cor. $H(X, Y|Z) = H(X|Z) + H(Y|X, Z)$.

$$- \sum_{x,y,z} p(x,y,z) \log p(x,y,z)$$

$$- \sum_{x,y,z} p(x,y,z) \log p(y|x,z)$$

$$- \sum_{x,z} p(x,z) \log p(x,z)$$

fix

2.3 Relative Entropy and mutual information.

(Def) Relative Entropy (Kullback-Leibler distance)

$$\Rightarrow D(p \parallel q) = \sum_x p(x) \log \left(\frac{p(x)}{q(x)} \right).$$

* $D(p \parallel q) \neq D(q \parallel p)$.

$\rightarrow (0 \log 0 = 0)$

Remark if $\exists x \in \mathcal{X}$ st $p(x) > 0, q(x) = 0$

$$\Rightarrow \boxed{D(p \parallel q) = \infty}$$
$$\left(p(x) \log \left(\frac{p(x)}{\underset{0}{q(x)}} \right) = \infty \right)$$

(Def) Mutual Information

$$\underline{I(X; Y)} = \sum_{x,y} p(x,y) \log \left(\frac{p(x,y)}{p(x)p(y)} \right)$$

(ex) if $X, Y \rightarrow$ independent $p(x,y) = p(x)p(y)$

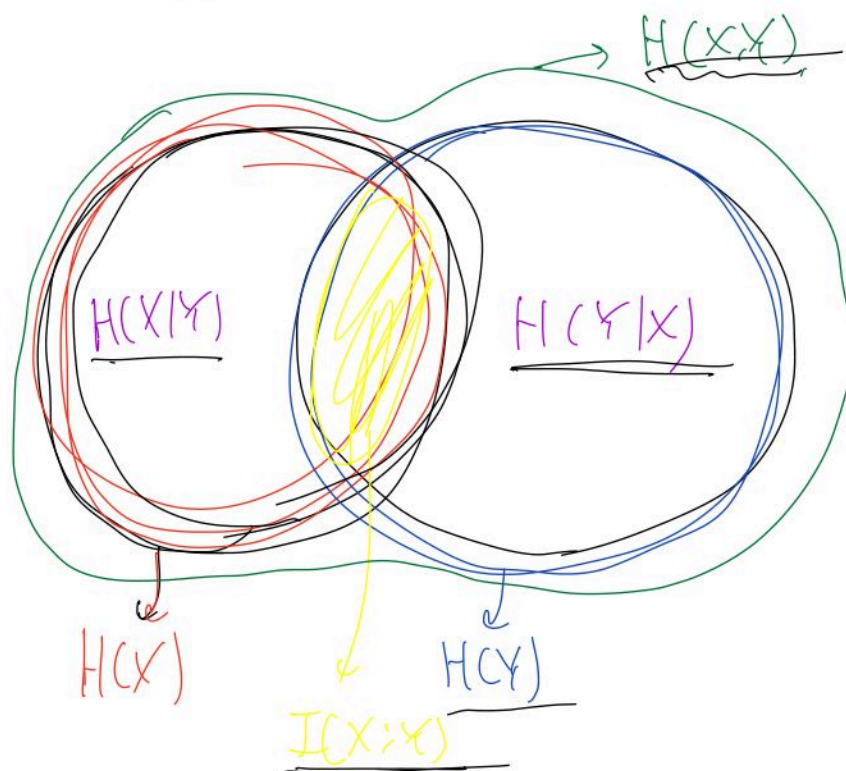
$$\Rightarrow \boxed{I(X; Y) = 0}.$$

2.4. Relationship between Entropy,

$$I(X; Y) = H(X) - H(X|Y),$$

(2/1) — . Omit. (2/2)

Thm 2.4.1 Mutual information and Entropy.



eg. $H(X, Y) = H(X|Y) + I(X; Y) + H(Y|X).$

° $H(X) = H(X|Y) + I(X; Y)$

2.5. Chain Rules for entropy, relative entropy.

Thm Chain Rule for Entropy

$$H(X_1, X_2, \dots, X_n) = \sum_{i=1}^n H(X_i | X_{i-1}, \dots, X_1).$$

(pf)

(conditional mutual information.)

Def $I(X; Y | Z) \doteq \sum_{x, y, z} p(x, y, z) \log \frac{p(x, y | z)}{p(x | z) p(y | z)}$

$$= [H(X | Z) - H(X | Y, Z)]$$

Thm Chain Rule for Information.

$$I(X_1, X_2, \dots, X_n; Y)$$

$$= \sum_{i=1}^n I(X_i; Y | X_{i-1}, \dots, X_1).$$

Def Conditional relative entropy.

$$D(p(y|x) || q(y|x)) = \sum_{y,x} p(x)p(y|x) \log \frac{p(y|x)}{q(y|x)}.$$

Thm Chain Rule for relative entropy.

$$D(p(x, y) || q(x, y)) = D(p(x) || q(x)) + D(p(y|x) || q(y|x)).$$

2.6. Jensen's inequality and its consequences.

Def ① convex ftn. : $f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda)f(x_2)$
 $\forall \lambda \in [0,1], x_1, x_2 \in X$

② strictly (.) : $["=" \Leftrightarrow \lambda = 0 \text{ or } 1]$

③ concave ftn : $-f$ is convex ftn
 $\Rightarrow f$ is concave ftn

Thm (Jensen's Ineq). f : convex.

$$* E(f(X)) \geq f(EX)$$

expectation value (p.p) omit. $\square$

Thm 2.6.3. (Information Inequality).

$$D(P \parallel Q) \geq 0 \quad (\text{equality : } P=Q)$$

$$(p.p) : D(P \parallel Q) = - \sum_{x \in A} p(x) \log \frac{p(x)}{q(x)}$$

$$X \supseteq A = \text{supp}(P)$$

(if $x \in A \Leftrightarrow p(x) \neq 0$).

$$= \sum_{x \in A} p(x) \log \left(\frac{q(x)}{p(x)} \right) \leq \log \left(\sum_{x \in A} p(x) \cdot \frac{q(x)}{p(x)} \right)$$

$$= \log \left(\sum_{x \in A} q(x) \right) \leq \log \left(\sum_{x \in \mathcal{X}} q(x) \right) = \log 1 = 0.$$

(A ⊆ X)

$$\boxed{D(p||q) \geq 0.}$$

Corollary ① $I(X;X) = D(p_{X,Y} || p_X p_Y) \geq 0.$

② $D(p_{Y|X} || q_{Y|X}) \geq 0.$

③ $I(X;Y(Z)) = D(p_{X,Y|Z} || p_X(Z) p_{Y|Z}) \geq 0.$

$$(\ast \ H(X) \geq 0)$$

Thm $H(X) \leq \log |\mathcal{X}|$ (the number of elements in the range of X)

(p.u.) $u(x) = 1/|\mathcal{X}|.$

$$D(p||u) = \sum_x p(x) \log \frac{p(x)}{u(x)}.$$

$$\begin{aligned} \underline{0 \leq} (J_{\text{rel. Ineq.}}) &= \sum_x p(x) (\log p(x) + \log |\mathcal{X}|) \\ &= -H(X) + 1 \cdot \log |\mathcal{X}|. \end{aligned}$$

$$\Rightarrow \boxed{H(X) \leq \log |\mathcal{X}|.}$$

Thus (Conditioning reduces entropy),

$$H(X|Y) \leq H(X)$$

(pf) $0 \leq I(X; Y) = H(X) - H(X|Y)$
 $\Rightarrow H(X) \geq H(X|Y)$ □

Thus (Independence bound on entropy),

$$H(X_1, X_2, \dots, X_n) \leq \sum_{i=1}^n H(X_i)$$

(pf) $H(X_1, \dots, X_n) = \sum_{i=1}^n H(X_i | X_1, \dots, X_{i-1})$
Chain rule $\leq \sum_{i=1}^n H(X_i)$ □

9:55 Begin.

2.7, Log sum Inequality and its application

Thus (Log sum Inequality)

$$a_1, \dots, a_n, b_1, \dots, b_n \geq 0$$

$$\Rightarrow \left(\sum_{i=1}^n a_i \log \frac{a_i}{b_i} \geq \sum_{i=1}^n a_i \log \left(\frac{\sum_{i=1}^n a_i}{\sum_{i=1}^n b_i} \right) \right)$$

(pf) (Key pt: Jensen's Ineq.)

Thus (Convexity of relative entropy)

$D(p||q)$ is convex in the pair (p, q) .

i.e. $D(\lambda p_1 + (1-\lambda)p_2 || \lambda q_1 + (1-\lambda)q_2) \leq \lambda D(p_1||q_1) + (1-\lambda)D(p_2||q_2)$.

(pf) L.H.S. = $\sum_x (\lambda p_1(x) + (1-\lambda)p_2(x)) \log \frac{\lambda p_1(x) + (1-\lambda)p_2(x)}{\lambda q_1(x) + (1-\lambda)q_2(x)}$

$$\leq \sum_x \left(\lambda p_1(x) \log \frac{\lambda p_1(x)}{\lambda q_1(x)} + (1-\lambda)p_2(x) \log \frac{(1-\lambda)p_2(x)}{(1-\lambda)q_2(x)} \right)$$

$$= \lambda D(p_1||q_1) + (1-\lambda)D(p_2||q_2) \quad \square$$

Thus $H(p)$ is a concave fcn of p .

(pf). $H(p) = \underbrace{\log |X|}_{\text{const}} - \underbrace{D(p||u)}_{\text{uniform}}$

$(p_1, u), (p_2, u)$.

$\square$

Thus (1) $I(X; X)$ is concave fcn of $p(x)$ for fixed $p_Y(x)$

(2) " " " " " "

2.8, Data-processing inequality

Def: X, Y, Z form a Markov chain $X \rightarrow Y \rightarrow Z$

$$\text{iff } p(x, y, z) = p(x) p(y|x) p(z|y)$$



Thm 2.8.1 (Data processing inequality) $X \rightarrow Y \rightarrow Z$
Markov chain
 $\Rightarrow I(X; Y) \geq I(X; Z)$

$$\begin{aligned} \text{pf } I(X; Y, Z) &= I(X; Z) + I(X; Y|Z) \\ &= I(X; Y) + I(X; Z|Y) \end{aligned}$$

$$I(X; Z|Y) = \sum_{y, z} p(y, z) \log \left(\frac{p(x, z|y)}{p(x|y)p(z|y)} \right)$$

$$I(X; Z|Y) = 0 \text{ if } (X, Z \text{ are indep}) = 0 \dots \dots = 0$$

$$I(X; Y) = I(X; Z) + I(X; Y|Z)$$

$$\Rightarrow I(X; Y) \geq I(X; Z)$$

Cor ① $Z = g(Y) \Rightarrow I(X; Y) \geq I(X; g(Y))$

② $I(X; Y) \geq I(X; Y|Z)$ $\{ I(X; g(g(Y)) = Y)$

2.9. Sufficient Statistics.

We consider $\{f_\theta(x)\}$: indexed by θ .

$$\theta \rightarrow X \rightarrow T(X)$$

any statistic (mean, variance, ...)

(Def) $\theta \rightarrow T$ is sufficient statistic relative to $\{f_\theta(x)\}$ if $\theta \rightarrow T(X) \rightarrow X$ forms a Markov chain

$\Rightarrow T$ is minimal sufficient statistic relative to $\{f_\theta(x)\}$ if $\theta \rightarrow T(X) \rightarrow U(X) \rightarrow X$ forms a Markov chain $\forall U$: sufficient statistic

(eg) $X_1, \dots, X_n, X_i \in \{0, 1\}$

i.i.d. (Independent & Identically distributed. ...)

$$\theta = \Pr(X_i = 1)$$

$$T(X_1, \dots, X_n) = \sum_{i=1}^n X_i$$

$$\Pr\left((X_1, \dots, X_n) = (\underline{x}_1, \dots, \underline{x}_n) \mid \sum_{i=1}^n X_i = k\right) = \begin{cases} 1/\binom{n}{k} & \sum x_i = k \\ 0 & \text{otherwise} \end{cases}$$

$\uparrow$

$$\theta \rightarrow \sum X_i \xrightarrow{\uparrow} (X_1, \dots, X_n)$$

$\rightarrow$ Independent θ

2.10. Fano's Ineq.

$$X \xrightarrow[\text{estimate}]{\hat{r}} Y \rightarrow \hat{X}.$$

Thm 2.10.1 (Fano's Inequality)

For any estimator $\hat{X}$ s. $X \rightarrow Y \rightarrow \hat{X}$.

with $P_e = \Pr(X \neq \hat{X})$, we have

$$H(P_e) + P_e \log |\mathcal{X}| \geq H(X|\hat{X}) \geq H(X|Y).$$

$$\text{Bmk } P_e = 0 \Leftrightarrow H(X|\hat{X}) = 0$$

(Same distribution)

Pr of Fano's Inequality

$$E = \begin{cases} 1 & \hat{X} \neq X \rightarrow P_e \\ 0 & \hat{X} = X \rightarrow (1 - P_e) \end{cases}$$

$$H(E, X|\hat{X}) = H(X|\hat{X}) + \underbrace{H(E|X, \hat{X})}_{\rightarrow 0}$$

(Chain) $\downarrow$

$$= H(E|\hat{X}) + H(X|E, \hat{X})$$

Conditioning
reduces
entropy

$$\leq H(P_e) + P_e \log |\mathcal{X}|$$

$$\begin{aligned}
 H(X|E, \tilde{X}) &= \underbrace{P(E=0)}_{\rightarrow} H(X|\tilde{X}, E=0) \\
 &\quad + \underbrace{P(E=1)}_{\rightarrow} H(X|E=1) \\
 &\leq (1-P_e) \cdot 0 + P_e \log |\mathcal{X}|
 \end{aligned}$$

$$\Rightarrow H(X|\tilde{X}) \leq H(E, X|\tilde{X}) \leq H(P_e) + P_e \log |\mathcal{X}|$$

$$\Rightarrow H(P_e) + P_e \log |\mathcal{X}| \geq H(X|\tilde{X}) \geq H(X|Y)$$



Car ① $\tilde{X} = Y \Rightarrow H(P_e) + P_e \log |\mathcal{X}| \geq H(X|Y)$

$$P = P_2(X \neq Y)$$

$$\textcircled{2} \quad H(P_e) + P_e \log(|\mathcal{X}|) \geq H(X|Y)$$

lem X, X' i.i.d with entropy $H(X)$.

$$P_r(X=X') \geq 2^{-H(X)}. \quad (\text{"} \Leftarrow \text{"} \Leftrightarrow X \text{ uniform})$$

(pf) $2^{E \log P} \leq E(2^{\log P}). \quad (\log \mathcal{X} = \log_2 \mathcal{X})$

$$\begin{aligned}
 \Rightarrow 2^{-H(X)} &\leq \sum_x p(x) 2^{\log p(x)} = \sum_{x \in \mathcal{X}} p(x)^2 \\
 &= P_r(X=X'). \quad \square
 \end{aligned}$$

Con. X, X' : Independent $X \sim p(x)$ $x, x' \in \mathcal{X}$
 $X' \sim r(x')$

$$\Rightarrow \begin{cases} P_1(X=X') \leq 2^{-H(p) - D(p||r)} \\ P_2(X=X') \leq 2^{-H(p) - D(p||r)} \end{cases}$$

$$\begin{aligned} (P_1) \quad 2^{-H(p) - D(p||r)} &= 2^{\sum (p \log p + p \log \frac{1}{r})} \\ &= 2^{\sum p \log r} \stackrel{?}{=} \sum p 2^{\log r} \\ &= \sum_x p(x) r(x) \\ &= P_2(X=X') \end{aligned}$$

QED